
To:	CPS Leadership & Stakeholders
From:	Rohit Jindal, Data Science — 1340 Project S.T.R.E.A.M.
Date:	June 2026
Re:	Proposed AI Capability: Making Sense of CIWP Data at District Scale

Hi,

Thank you for sharing the two ideas — I have read through both carefully and want to say that this is a genuinely exciting problem to work on. Not because it is technically interesting (though it is), but because the gap it describes is real: 570 schools are putting serious thought into their improvement plans every year, and right now there is no practical way for the district to hear all of them at once. The current process requires someone to read manually, and at 570 plans, that simply does not scale.

I believe this is something we can build as a lasting district capability — not a one-time analysis, but a tool that gets used every cycle. Below I have outlined three approaches, from the most immediate to the most comprehensive. For each one, I have described what it does in plain terms, what you can expect from it, and what it would cost to run.

I have also added optional technical detail boxes for each option — you do not need to read them, but they are there if anyone wants to understand the mechanics.

Option 1 — Automated Trend Mining Across All 570 CIWPs

The problem it solves

There is already a tool — the Gem — that works well when a network leader uploads a small batch of plans, say 20 to 30. The problem is that it was not designed for district-wide scale. When you try to run it across all 570 CIWPs, the results become inconsistent and unreliable. So right now, district-level trend analysis either does not happen or happens through time-consuming manual effort.

What I would build

A process that reads each CIWP individually, pulls out the four key pieces — the Student-Centered Problem, Root Cause, Theory of Action, and Practice Goals — and stores them in a clean, organized database. Once all 570 are processed, a second step looks across the entire database to count patterns, group similar themes, and produce a plain-language summary report.

The output would look something like: "The most commonly cited root cause across district elementary schools is X, appearing in 148 plans. The most common associated strategy is Y."

And it would be filterable — a math director could see only the plans relevant to math, an early childhood director only theirs.



Optional Technical Detail

Each CIWP is sent to a Large Language Model (LLM) — the same family of AI that powers tools like ChatGPT — with a structured prompt that asks it to extract specific fields consistently.

The extracted data is stored row by row in a database: one row per school per CIWP cycle.

A second AI pass runs across the aggregated database to surface themes, using a technique called topic modeling — it finds clusters of similar language without being told in advance what to look for.

This sidesteps the limitation of the current Gem by never asking the AI to read 570 documents at once. Each document is handled individually, then combined at the database level.

What to expect

- A clean, structured view of all 570 CIWPs — problems, root causes, strategies, and goals — in one place.
- Filterable trend reports by network, grade band, or subject area.
- The process runs on demand and can be repeated every cycle without starting from scratch.
- Accuracy for structured extraction from well-written plans is typically in the range of 85–90%. A human spot-check on a sample is advisable before sharing results with leadership.

What it costs to run

Processing all 570 CIWPs through an AI service (I would use a cost-optimized model designed for exactly this type of extraction work) costs roughly \$15 to \$40 per district-wide run, depending on how long the plans are. For an annual cycle, total AI processing costs would be under \$100. There is no recurring infrastructure cost — the process runs on demand.

Option 2 — A Search Tool: Ask a Question, Get an Answer from the Data

The problem it solves

A periodic trend report answers broad questions well. But CO directors often have very specific questions that a fixed report cannot answer — for example: "What are schools in Network 14 saying about family engagement as a root cause?" or "Which schools have a Theory of Action involving teacher coaching, and what metrics are they using?" Right now, answering those questions means reading plans manually.

What I would build

On top of Option 1, I would build a simple tool — accessible through a browser — where any CO director can type a question in plain English and receive an answer drawn directly from the CIWP data. The answer would not be a guess. Every response would be traceable back to the actual plans that support it, with the school name and the relevant text cited.

Think of it as giving district leadership a direct line to what schools are actually saying — without having to read a single plan yourself.



Optional Technical Detail

This is built using a technique called RAG — Retrieval Augmented Generation. In plain terms: the AI is given access to a well-organized library of CIWP excerpts, and is only allowed to answer based on what is in that library. It cannot make things up.

Each section of each CIWP is converted into a numerical fingerprint (called an embedding) that captures its meaning. When a question is asked, the system finds the most relevant excerpts using those fingerprints, then synthesizes a grounded answer.

This means the system will say "I do not have enough data to answer that" rather than fabricate a response — which is the key difference from a general-purpose chatbot.

Built on top of Option 1 — Option 1 is a required foundation.

What to expect

- Any CO director can query the CIWP corpus in plain English, without any technical knowledge.
- Answers include source attribution — which schools, which plans, which section.
- Filterable by network, grade band, subject area, or CIWP cycle year.
- Updated annually as new CIWPs are processed — no rebuilding from scratch each time.

What it costs to run

Generating the underlying index from 570 CIWPs is a one-time cost per cycle — roughly \$5 to \$20. Each director query costs fractions of a cent. For a typical usage pattern of a few hundred queries a month, total ongoing AI cost would be under \$15 per month. The main investment here is in the initial build.

Option 3 — Informing Future CIWP Cycles with Historical Intelligence

The problem it solves

The second idea in the document is about the longer horizon: helping CO leaders set smarter CIWP Foundations every three years, and helping school teams do faster, more comprehensive self-assessments so they spend less time on data and more time on the conversations that matter. This is the most ambitious of the three — and the most transformative.

What I would build

Building on Options 1 and 2, this layer adds historical depth. Over time, the system would show not just what schools are prioritizing now, but how those priorities have shifted across cycles — and where there is a mismatch between what schools say they are working on and what their outcome data suggests they should be working on.

For school CIWP teams specifically, I could build a personalized briefing tool: a school provides their performance data, and the system produces a plain-language summary of their key

challenges — the kind of briefing that currently takes a school team hours to pull together before they can even start their reflection conversation.

 Optional Technical Detail

Multi-cycle CIWP data is stored in a longitudinal database, enabling year-over-year trend comparisons. Unsupervised topic modeling surfaces themes that emerge across cycles without needing predefined categories. School-level outcome data (attendance, assessment, etc.) is joined to CIWP data, allowing the system to flag when a school's stated priorities and their measured challenges are misaligned. An automated narrative generation step produces a plain-language performance briefing for each school's CIWP team, using their structured data as input to an LLM.

What to expect

- CO leaders set CIWP Foundations backed by actual multi-year district trend data, not anecdote.
- School teams spend significantly less time parsing data and more time on stakeholder engagement — directly addressing the bottleneck the proposal identifies.
- A living district intelligence layer that improves in value every cycle as more data accumulates.

What it costs to run

Ongoing AI processing costs for this phase remain low — under \$200 per year. The more meaningful investment is in building and maintaining the infrastructure: a persistent database, scheduled pipelines, and the web application. This is the phase where having additional data science capacity would make the most material difference to both quality and scope.

Summary: The Three Options at a Glance

Option	What It Does	Annual Run Cost	Scope
Option 1 — Trend Mining	Reads all 570 CIWPs, extracts structured data, produces filterable trend reports	< \$100	Immediate value, Spring/Summer 2026
Option 2 — Search & Query	Plain-English question tool for CO directors, answers grounded in actual CIWP text	< \$150	Ongoing CO planning and alignment
Option 3 — Historical Intelligence	Cross-cycle analysis, school-level data briefings, informs future CIWP Foundation-setting	< \$200	SY30–32 CIWP Foundation planning

What This Means for CPS — and How I Would Like to Proceed

The efficiency picture

If each of 570 CIWPs takes a person just 30 minutes to read and summarize into a usable format, that is 285 hours of work — before any analysis, aggregation, or reporting. Realistically, doing this properly is weeks of analyst time, every single year, just to produce what Option 1 would generate in a single afternoon.

But beyond the time savings, the more important gain is completeness. A person reading under deadline pressure will miss things. An automated system applied consistently will not. Every school's plan gets heard, every cycle, at the same level of attention.

How I would like to proceed

I am genuinely excited about this and would love to explore it further with the right stakeholders. These three options are not competing proposals — they are a natural progression, with each one building on the last. The right starting point depends on what CPS needs most urgently and what resourcing is available.

I should also be transparent: this is ambitious work, and the more comprehensive the capability, the more it benefits from dedicated data science capacity. A broader team would not just accelerate delivery — it would produce something more robust, better validated, and more likely to be trusted and adopted at scale. I am happy to discuss what that could look like.

I would welcome the opportunity to walk through any of this in more detail. Please feel free to reach out — I am available for a conversation whenever it is convenient.

Best regards,

Rohit Jindal (Senior Data Scientist)

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